Candidate Seat no: _________

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

B. Tech. EE Sem-V Theory Examination

(March -2018)

EE 303.01/303 Microcontrollers & Applications

Date: 31/03/2018, Saturday **Time: 1:30 p.m. to 4:30 p.m.** **Maximum Marks: 70**

SECTION-1

Q-1 Draw internal architecture of 8051. Explain function of each block in brief. **[8]**

Q-2(a) Draw port structure of Port 0 and Port 2, also explain why we have to connect extern pull up register to port 0 when using it as an output port. **[7]**

OR

Q-2(a) Explain all mode of serial communication. **[7]**

Q-2(b) Draw and Explain the Pin diagram for 8051 microcontroller with clear description of

function of each pin. **[7]**

OR

Q-2(b) Explain the interfacing of single key and associated problem and solution with required diagrams. **[7]**

Q-3(a) Draw and explain the internal diagram of Timer in mode 1 and mode 2 of operation. **[7]**

OR

Q-3(a) Explain both software key debouncing (with flowchart) and Hardware key debouncing (with circuit diagram). **[7]**

Q-3(b) **Attempt any three.** **[6]**

1. Explain difference between Microcontroller 8051 and Microprocessor 8085.
2. Explain difference between Assembly level language and Embedded C language.
3. Find out binary and hexadecimal number of $(25)_{10}$.
4. Upon hardware reset content of PC=____ and content of SP=_____.
5. Explain simplex, half-duplex and duplex serial communication.

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SECTION-2

Q-4(a) Calculate number to be loaded in timer registers to generate time delay of 50milisecond and Write C program to toggle status of P1.0 after each 50milisecond delay using timer module. (Consider crystal frequency of 12Mhz.) **[7]**

Q-4(b) Explain function of all pins of 16x2 LCD module and draw interfacing diagram. **[7]**

OR

Q-4(b) Draw interfacing diagram of ADC0809 and explain pin function of each pin of ADC. **[7]**

Q-5(a) Draw and explain function of register listed below. (attempt any four) **[12]**

1. PSW
2. TMOD
3. TCON
4. IE
5. SCON
6. PCON
7. IP

Q-5(b) Write C program (attempt any one) **[2]**

1. To toggle status of port 0 continuously with some time delay.
2. To send number 0 to 9 through port 0 with some time delay.

Q-6(a) Write C program (attempt any one) **[3]**

1. To toggle status of pin P1.7 each time falling edge appeare at pin P3.2.
2. To turn on LED connected to pin P1.7 when status of pin P1.0 is high and turn LED off if status of port pin P1.0 is low.

Q-6(b) Write C program (attempt any one) **[4]**

1. To print "CHARUSAT" on 16x2 LCD. (draw interfacing diagram to justify program)
2. To send "Electrical Engineering" continuously through serial port at baud rate of 9600.

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